INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

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6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives.

6.1 Answers to the research questions

The thesis built a coupled PageRank (C-PR), one walk over the ERC-20 allowance and transfer edges between Arbitrum One contracts, and set it beside EndorseRank, Adaptive Weighted PageRank (AWP) (T. Do et al., 2023), the ablation S-PR and the raw counts in a discovery-order test at two freezes and in the holdout windows W0 (April to June 2026) and W1 (July to September 2026). After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. The verdicts below under rules A and B are a re-analysis, and the registered analysis gave the same readings (Appendix 8.6). Figure 6.1 sets the five answers side by side.

RQ1 asked whether EndorseRank and AWP order the same contracts differently, whether each agrees most with current checks of its own edge type, and what each walk costs. H1 is supported, largely by construction: most cohort contracts hold no capped allowance from another contract and score zero under EndorseRank, so both walks order only about a third of the contract pairs (Section 4.4). Each walk agrees most with the proxies of its own edges (Table 4.5), which supports H2 only narrowly, since the proxies come from the same events as the graphs. Cost follows the size of the layers walked and the iterations needed, so EndorseRank, on the far smaller allowance layer, is cheapest and C-PR takes slightly longer than AWP (Table 4.3); H3 is supported, largely

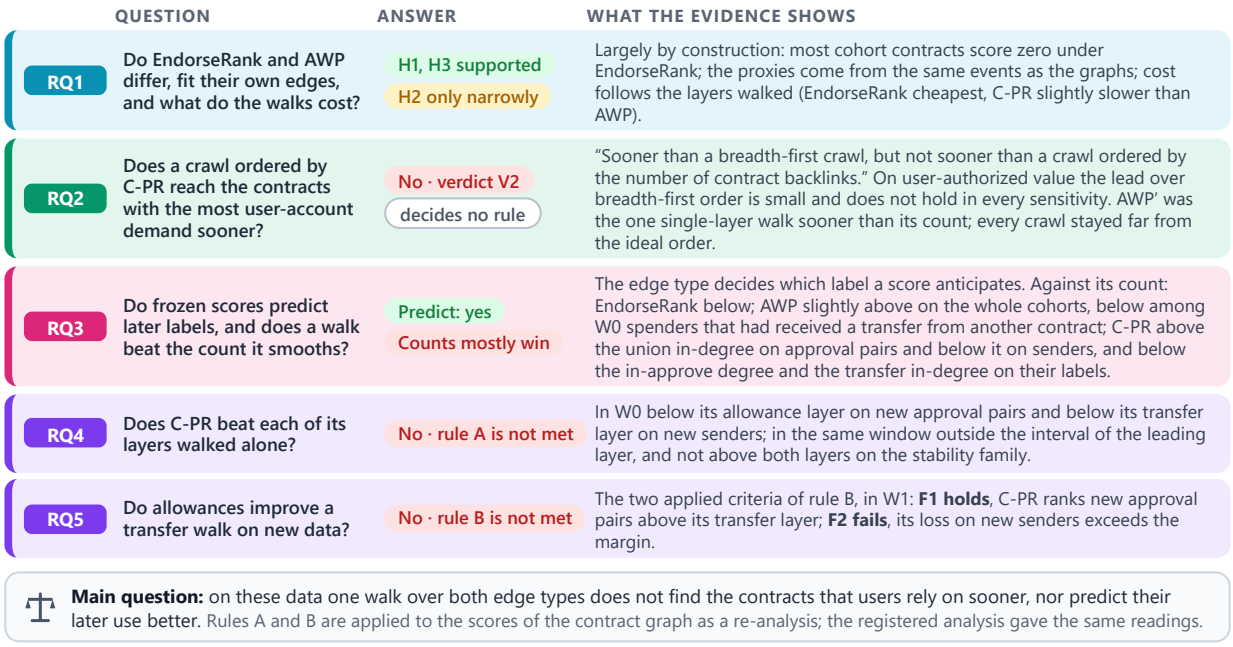


Figure 6.1: The answers to RQ1–RQ5, in the words of this section. The quotation is the verdict sentence of the discovery-order test, which was defined after the registered results were known and decides no rule of the plan or the registration. The thesis applies two of rule B's registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. Rules A and B are applied to the scores of the contract graph as a re-analysis.

by construction.

RQ2 asked whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than breadth-first and backlink-count crawls. The answer is no. The test's decision rule, read at T_{obs} in the primary reveal mode, gives verdict V2 on both importances (Table 4.12): "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." On user-authorized value its lead over breadth-first order is small and does not hold in every sensitivity (Table 8.9). Outside the decision (Table 4.14), AWP′ was the one single-layer walk sooner than its count on both importances, and the allowance layer added a small, resolved amount to C-PR′ over its transfer layer. In the other reveal

mode, which shows out-edges only and carries no verdict, C-PR′ was sooner than both comparators at both freezes. Every crawl stayed far from the ideal order (Table 4.13). The test decides no rule of the plan or the registration.

RQ3 asked whether frozen scores predict later new approval pairs and new transfer senders, and whether any walk beats the raw count of the edges it smooths. The first part holds in both windows, and the edge type decides which label a score anticipates (Tables 4.15 and 4.20). On the second the counts mostly win. EndorseRank ranks later approval pairs below the in-approve degree in both windows with either restart rule (Section 4.7; Tables 4.16 and 4.18). AWP ranks later senders slightly above the transfer in-degree on the whole cohorts, but below it among the W0 spenders that had received a transfer from another contract before the freeze (Table 4.17). C-PR ranks later approval pairs above the union in-degree, the count of its edges, and later senders below it, and lies below the in-approve degree and the transfer in-degree on their labels (Table 4.22). The labels count mostly user accounts, which no score reads, and more than a quarter of each spender cohort lies outside the union graph and scores zero (Tables 8.6 and 4.2).

RQ4 asked whether the coupled walk ranks contracts better than either of its layers walked alone, in and out of window. The answer is no. Rule A is not met in either part: in W0 C-PR ranks new approval pairs below its allowance layer and new senders below its transfer layer (Table 4.16), and in the same window it lies outside the interval of the leading layer on the transfer and allowance families and does not exceed both layers on the stability family (Table 4.19). Among the spenders inside the union graph the contrast with the allowance layer does not exclude zero, a comparison that decides no rule (Table 4.23).

RQ5 asked whether adding the allowance layer improves a transfer walk on new data, gaining on later approvals without losing on later inflow. The answer is again no. The thesis applies two of rule B's registered criteria, F1 and F2; under the registration the rule holds only if each of its criteria holds. F1 holds: in W1 C-PR ranks new approval pairs above its transfer layer walked alone. F2 fails: its loss on new senders exceeds the margin. Rule B is not met, and the two registered sensitivity analyses give the same reading (Table 4.21).

Claims C1–C6 (Section 4.9) state these findings formally.

The main research question asked whether one walk over both edge types finds the contracts that users rely on sooner, and predicts their later use better, than single-layer walks and raw counts. On these data it does not. Its crawl is sooner than breadth-first but later than the backlink-count and AWP′ crawls, it meets neither rule A nor rule B, and out of window it beats neither single-layer count on that count's own label. What the allowance layer adds is small and resolved: a little in discovery order, and a gain on later approval pairs that comes with a loss on later inflow.

6.2 Original contribution to knowledge

Read against the literature of Chapter 2, the five contributions of Section 1.9 establish the following. In the work reviewed, graph reputation on public ledgers ranked addresses by the transfers among all of them (T. Do et al., 2023), by risk propagation in transaction networks (D. Lin et al., 2025) or by explicit ratings (Bellini et al., 2020), and DeFi reputation tooling scored an address's holder from credentials (Human Passport, n.d.). None of it ranked contracts on the relations among contracts alone, used the ERC-20 allowance (Vogelsteller & Buterin, 2015) as the edge of a ranking or coupled it with transfers in one walk; approvals had been used to cluster the addresses of one entity (Victor, 2020) and studied as a security risk (Wang et al., 2022). None asked of a ledger ranking what the crawl-ordering experiment asked of web rankings (Cho et al., 1998), how soon it leads to what is important, and none set on-chain scores beside the raw counts they smooth, with confidence intervals, paired differences and temporal holdouts. This thesis supplies these.

The new knowledge has five parts. First, a ranking of contracts can be computed from the relations among contracts alone, and what it measures depends on its edges, which can be checked on one cohort with quantified uncertainty. Second, the coupled walk leads a crawl to the contracts users rely on sooner than breadth-first order but not sooner than the count of contract backlinks, a result that depends on what a visit reveals. Third, the coupled walk meets neither rule that sets it against its own layers. Fourth, the counts are hard to beat: C-PR ranks no label above the degree of that

label's edge type, and of the single-layer walks only AWP leads its count, in the deciding discovery-order run on both importances and, slightly, on later inflow, though not among the spenders that had received a transfer from another contract. Fifth, an allowance between contracts is rarely backed: counted up to the owner's balance, most such allowances form no edge, and on the thin layer that remains a walk with uniform restarts is at depth one, up to a constant, a linear function of the transition mass into a spender (Section 5.3.2). The open pipeline, whose published summaries let others recompute every table, is part of the contribution, since automated scores that outsiders cannot inspect are a known concern (Packin & Lev-Aretz, 2024).

The scope is narrow. The results apply to the contract graph of the five tokens on Arbitrum One in these months, with addresses typed from code read at one block, to the cohorts drawn from it and to the coupling the plan fixed (a fixed λ, dollar-weighted layers, uniform restarts). The allowance layer between contracts is thin, and at T_{obs} one owner contract holds most of its pairs, so that contract's allowances weigh on the allowance-side results of that freeze (Table 4.1). No score reads a contract's code or says whether it is safe.

6.3 Future work

Each suggestion below answers a limitation of Chapter 5. Any new score, weighting or test it leads to should be fixed before it is scored, replicated on new data and checked against the counts it smooths.

- Credit decisions about contracts. A ranking that found the contracts users rely on could serve as one input to such decisions. The thesis does not test this use, and its results do not yet support it: the walk did not find those contracts sooner than a count, and the use would also need a contract outcome bearing on repayment, which this study lacked.
- Importance labels from outside the chain. The discovery-order test measures importance on chain, as the capped dollars user accounts transfer to a contract or let it spend, and so depends on the rule that separates user accounts from contracts. Labels of which contracts

matter, collected off chain by people who examine contracts, would test the orderings against a judgment sharing no data with them.

- Detection of smart accounts. The rule recognizes Safes and ERC-4337 accounts. The kinds it misses, and contracts that services create for a single user, order or payment, stay in the graphs as contracts (Appendix 8.4.3); a better rule would show how far they shape the transfer layer and the importance measures.
- A later freeze. The discovery-order test and the comparisons with the counts were defined after the registered results were known, and the two freezes share nine months of their score windows. Repeating the test on a later freeze, with its protocol and primary reveal mode fixed before that freeze's data exist, would give a registered answer to RQ2, and other rollups would show whether the answer extends beyond Arbitrum One.
- A coupled walk with activity restarts. C-PR differs from AWP in coupling, weights and restarts at once, and AWP′, which restarts by sending activity, was the soonest walk of the deciding run. A coupled walk whose layers carry the edges and weights of AWP and EndorseRank and whose restarts mix sending and approving activity (C-AWP) would set the coupling alone against the published walk.

6.4 Closing reflection

A rollup's contracts are many, and the relations among them are public. A transfer between two contracts records that one moved tokens to the other, and an allowance, counted only up to what its owner holds, records a spending right the owner can back. PageRank will rank either graph, and one walk can follow both. Here that walk was not the best guide to the contracts that users rely on. A crawl ordered by the number of contracts that send transfers to or approve a contract reached them sooner, and out of window each edge type's count ranked the later arrivals of that type above the coupled walk. For a wallet application, an explorer or an auditor choosing from public logs which contracts to examine first, the counts on the contract graph are the baseline a walk must beat:

they cost least, and here they beat it. A walk earns its cost only where shown to beat its count, which AWP did only in part. The allowance between contracts proved a thin layer. Whether a ranking built from these relations finds what people who examine contracts would choose is a question for labels such people supply.